

Spencer Vaughan

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Summary

Computing & Software Systems graduate with over a decade of experience in competitive robotics, autonomous systems and STEM outreach. Proven leadership as founding President and Team Lead of a national championship-winning university robotics team, leading cross-functional teams and managing six-figure budgets. Strong background in software engineering, real-time systems, and hardware–software integration. Commencing a part-time Master of Software Engineering (Artificial Intelligence) in July 2026. Seeking graduate software engineering and robotics roles, and opportunities to engage with industry to further develop technical, systems, and professional skills.

Education

Master of Software Engineering (Artificial Intelligence) Jul 2026 – Jun 2029
University of Melbourne

- Commencing July 2026, part-time, after-hours study
- Planning to major in Artificial Intelligence

Bachelor of Science (Computing & Software Systems) Jan 2021 – Dec 2025
University of Melbourne

- Completed subjects in Artificial Intelligence, Machine Learning, and Object Oriented Software Development, with hands-on experience using LLMs to drive intelligent outputs and support problem-solving
- Capstone project focused on full-stack web development, applying SCRUM and Agile methodologies with regular client meetings to gather requirements, iterate on feedback and deliver a tailored solution
- Completion of Melbourne Plus: People Leadership Certification

Relevant Experience

President & Team Lead Feb 2021 – Dec 2024
Robotics, Engineering & Competition Society (EMU5), Parkville

- Led a 40+ member interdisciplinary robotics team, managing six-figure annual budget and steering team to 3× University National Championships (2021–2024) and 3× National Excellence Awards
- Founded RECS & Team EMU5, initiating development projects in motion profiling, PID tuning, real-time C++ control systems, and sensor fusion
- Orchestrated collaboration between software, electrical, and mechanical teams to develop cohesive system designs, resulting in reliable, high-performance prototypes
- Oversaw CAD review, mechanical implementation, and housing design for sensors and electronics
- Managed logistics for national and international competitions including flights, accommodation, freight, and risk assessments
- Established project management framework using Gantt charts, structured meeting cycles, and milestone tracking
- Recognised with the Innovate Award (2021) and Create Award (2023) at the VEX Robotics World Championships

Regional Support Specialist

Jan 2023 – Present

Robotics Education & Competition Foundation, Australia

- Provided on-site technical and logistical support at dozens of robotics competitions, troubleshooting hardware and software issues
- Led professional development workshops for 50+ educators across Victoria on STEM pedagogy and robotics curriculum integration
- Built strong relationships with students, educators, and industry partners as a mentor and technical advisor
- Maintained competition integrity by training teams and volunteers on rules, judging, and event systems
- Delivered operations at 7 championship-level robotics events across Australia, USA, and Thailand
- Fostered inclusive environments emphasising real-world STEM applications and critical thinking

Secretary & Workshop Coordinator

Jan 2021 – Present

in2robotics, Australia

- Volunteered with a not-for-profit charity expanding access to STEM education
- Designed and delivered robotics workshops for Year 7–12 students
- Mentored under-represented students and supported outreach programs

Shift Manager

Dec 2020 – Present

McDonalds, North-Western Suburbs

- Worked across 8+ store locations, adapting quickly to new systems and procedures
- Completed the Operations Technology Program exploring technology to enhance sales and efficiency
- Made independent operational decisions during overnight store management

Technical Skills

Programming Languages: Java, C++, C, JavaScript, Python, Rust, LaTeX, SQL

Software Development & Methodologies: Object Oriented Programming, Software Development Lifecycle (SDLC), Agile/SCRUM, Version Control, Data Structures & Algorithms

Technical Tools & Documentation: Databases, Azure, Office Suite, Technical Documentation

Autonomous Systems & Robotics: Real-Time Systems, Microcontrollers, PCB Design, Sensor Fusion, Motion Profiling, PID

Key Skills

- Project & Time Management
- Critical Thinking
- Innovation & Systems Thinking
- Collaboration & Cross-functional Teamwork
- Communication & Public Speaking
- Leadership & Mentoring